

MMIS 885: Management Information System

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Lecture 3:MMIS 885

Building Information System (System Analysis and Design)

Lecture 3: Outline

- Introduction
- Organizational Transformation enabled by IT
- Business Process Management
- Overview of System Development
- Methods for Modelling and Designing Systems
- Other methods
- Application Development for Digital firm

Organizational Transformation Enabled by Information Technology (IT)

- Organizations adopt information systems not only to automate existing tasks but also to transform how they operate and compete fundamentally.
- Information technology enables four progressive levels of organizational change, ranging from simple automation to complete business transformation.
- Modern organizations often progress through these stages, evolving from operational efficiency to strategic innovation and competitive advantage.

Organizational Transformation Enabled by IT

- **1. Automation**
 - Automation involves using information systems and digital technologies to perform routine, repetitive, and rule-based tasks with minimal human intervention.
- **Objectives**
 - Increase operational efficiency, faster processing, improve accuracy, and minimize human errors
- **Examples**
 - Payroll processing systems, Automated inventory management, Electronic attendance systems, and Chatbots for customer service
- **Benefits**
 - Faster operations, Lower labor costs, Improved productivity, and Consistent service delivery
- **Modern Trend**
 - Organizations increasingly use **Artificial Intelligence (AI)** and **Robotic Process Automation (RPA)** to automate not only routine tasks but also cognitive tasks such as document processing and customer support.

Organizational Transformation Enabled by IT

- **2. Business Process Improvement (Process Rationalization)**
 - Business Process Improvement (BPI) involves examining existing business procedures to eliminate inefficiencies, redundancies, delays, and unnecessary activities.
 - streamlines and optimizes existing business processes.
- **Objectives**
 - Simplify workflows, Eliminate bottlenecks, Standardize procedures
 - Improve service quality, Reduce waste
- **Common Improvement Methodologies**
 - **Total Quality Management (TQM)**
 - A continuous improvement philosophy that emphasizes customer satisfaction through organization-wide quality improvement.
 - Six sigma
 - A data-driven methodology aimed at reducing process variation and defects.
- **Modern Examples**
 - Electronic procurement systems and Online patient appointment scheduling

Organizational Transformation Enabled by IT

- **3. Business Process Redesign (BPR)**
 - Business Process Redesign (or Business Process Reengineering) involves fundamentally rethinking and radically redesigning business processes to achieve dramatic performance improvements.
 - Rather than making small improvements, BPR asks:
 - *"If we were starting this organization today, how would we design the process?"*
- **Characteristics**
 - Radical rather than incremental change
 - End-to-end process redesign
 - Elimination of unnecessary steps
 - Cross-functional collaboration
 - Heavy use of digital technologies

Organizational Transformation Enabled by IT

- **Activities**
 - Analyze existing ("As-Is") processes
 - Identify non-value-added activities
 - Simplify workflows
 - Integrate fragmented systems
 - Automate redesigned processes
- **Example**
 - **Traditional Loan Approval**
 - Customer → Clerk → Supervisor → Manager → Credit Officer → Finance → Approval
 - **Redesigned Digital Process**
 - Customer Portal → AI Credit Assessment → Automated Approval → Instant Notification
- **Benefits**
 - Shorter processing times
 - Lower operational costs
 - Improved customer experience
 - Increased organizational agility

Organizational Transformation Enabled by IT

- **4. Paradigm Shift (Digital Transformation)**
 - A paradigm shift occurs when information technology fundamentally changes the organization's business model, products, services, or industry position.
 - Rather than improving existing operations, organizations reinvent themselves through digital innovation.
- **Characteristics**
 - New revenue models
 - Digital-first operations
 - Platform-based business models
 - Data-driven decision making
 - AI-enabled services

Organizational Transformation Enabled by IT

- Examples of Paradigm Shift

Traditional Organization	Digital Transformation (Paradigm Shift)
Physical bank branches	Mobile banking and digital wallets
Classroom-only teaching	Hybrid and online learning platforms
Taxi companies	Ride-hailing platforms (Uber, Bolt)
DVD rentals	Netflix streaming
Physical retail	Amazon and e-commerce marketplaces

Organizational Transformation Enabled by IT

- Comparison of organizational changes enabled by IT

Level	Scope of Change	Risk/Return	Examples
Automation	Individual tasks	Low/Low	Payroll, attendance systems
Business Process Improvement	Existing business processes	Moderate/Moderate	Workflow optimization
Business Process Redesign	Entire business processes	High/High	ERP implementation, digital banking
Paradigm Shift	Entire organization/business model	Very High/Very High	Amazon, Netflix, Uber, AI-driven enterprises

Business Process Management (BPM)

- **What is BPM?**
 - Business Process Management (BPM) is a structured approach to **analyzing, designing, implementing, monitoring, and continuously improving** business processes to enhance organizational performance.
- **Objectives of BPM**
 - Improve operational efficiency and productivity
 - Reduce costs and process cycle time
 - Enhance customer satisfaction
 - Ensure compliance with business policies and regulations
 - Support digital transformation and organizational agility
- **The BPM Lifecycle**
 - **Identify** processes that require improvement.
 - **Analyze** the current (As-Is) process.
 - **Design** an improved (To-Be) process.
 - **Implement** the redesigned process.
 - **Monitor and continuously improve** using performance metrics (KPIs).

Business Process Management (BPM)

- **Key BPM Capabilities**
- Modern BPM solutions help organizations to:
 - **Document and map** existing business processes
 - **Identify bottlenecks** and process inefficiencies
 - **Model and redesign** improved workflows
 - **Automate business processes** using workflow automation and Robotic Process Automation (RPA)
 - **Manage business rules** to ensure consistency and compliance
 - **Monitor performance** through dashboards and analytics
- **Emerging BPM Technologies**
 - Artificial Intelligence (AI)
 - Process Mining
 - Robotic Process Automation (RPA)
 - Cloud-based BPM Platforms
 - Business Intelligence (BI) Dashboards

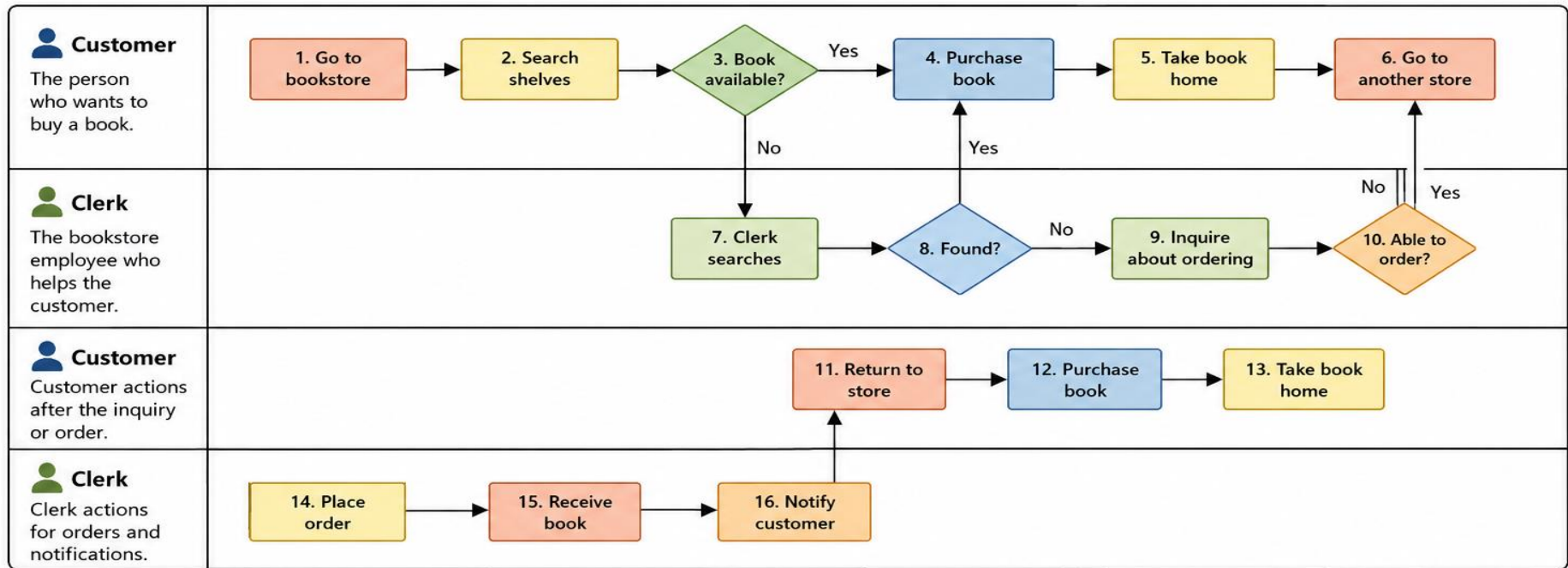
Benefits and Business Value of BPM

- **Benefits of BPM**
 - Faster and more efficient business processes
 - Lower operational costs
 - Improved quality and accuracy
 - Better customer experience
 - Increased organizational agility
 - Data-driven decision making
 - Continuous process improvement
- **Example 1: University Student Registration**
 - **Before BPM**
 - Manual registration
 - Long queues
 - Paper-based approvals
 - Frequent errors
 - **After BPM**
 - Online registration portal
 - Automated approvals
 - Real-time status tracking
 - Faster service delivery

Example 2: Purchasing a Book

AS-IS BUSINESS PROCESS FOR PURCHASING A BOOK FROM A PHYSICAL BOOKSTORE

This diagram shows the current (as-is) process of how a customer purchases a book in a physical bookstore.



STEP-BY-STEP EXPLANATION

- 1 Customer goes to the bookstore.
- 2 Customer searches the shelves for the desired book.
- 3 Decision: Is the book available on the shelf?
- 4 If yes, customer purchases the book.
- 5 Customer takes the book home.

- 6 If not available, customer may go to another store.
- 7 Clerk searches for the book (in store or backroom).
- 8 Decision: Is the book found?
- 9 If not found, clerk inquires if the book can be ordered.
- 10 Decision: Is the book able to be ordered?

- 11 If orderable, customer returns to the store.
- 12 Customer purchases the book when it arrives.
- 13 Customer takes the book home.
- 14 Clerk places an order for the book.
- 15 Clerk receives the book from the supplier.
- 16 Clerk notifies the customer that the book is ready.

LEGEND

- Customer action
- Clerk action
- Purchase/payment action
- Take home / result
- Decision point
- Flow of process

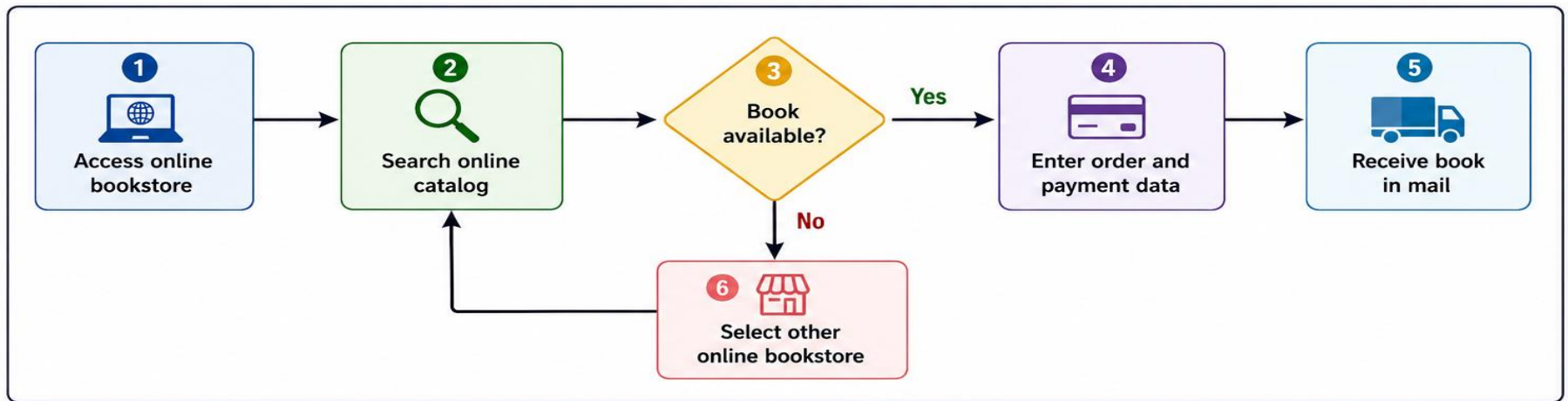


KEY TAKEAWAY: Purchasing a book from a physical bookstore involves multiple interactions between the customer and the clerk, including searching, checking availability, ordering (if needed), and final purchase.

Example 2: Purchasing a Book

REDESIGNED PROCESS FOR PURCHASING A BOOK ONLINE

Using Internet technology streamlines the book purchasing process, reduces steps, and requires fewer resources.



STEP-BY-STEP EXPLANATION

- 1 Access online bookstore:** The customer visits an online bookstore using the Internet.
- 2 Search online catalog:** The customer searches the online catalog for the desired book.
- 3 Book available?:** The system checks if the book is available.
 - **Yes:** Continue to place the order.
 - **No:** Select a different online bookstore and search again.
- 4 Enter order and payment data:** The customer enters shipping information and makes the payment online.
- 5 Receive book in mail:** The bookstore processes the order and delivers the book to the customer's address.
- 6 Select other online bookstore:** If the book is not available, the customer chooses another online bookstore and repeats the search.

PROCESS SUMMARY



The online book purchasing process consists of five main steps when the book is available. If the book is not available, the customer can easily choose another online bookstore and search again. This redesigned process is faster, more convenient, and uses fewer resources compared to the traditional in-store process.

LEGEND

- Customer / System Action
- Decision Point
- Alternative Action (If No)
- Flow of Process
- Yes** Positive Decision Path
- No** Negative Decision Path
- 1** Step Number

System Development

- **What is Systems Development?**
 - **Systems Development** is the structured process of creating, implementing, and maintaining an information system to address an organization's business problems, opportunities, or strategic objectives.
 - It involves a series of interconnected activities that ensure the system meets user requirements and delivers business value.
- **Major Activities in Systems Development**
 - **Systems Analysis**
 - **Systems Design**
 - **Programming (Development)**
 - **Testing**
 - **Implementation (System Conversion)**
 - **Production and Maintenance**

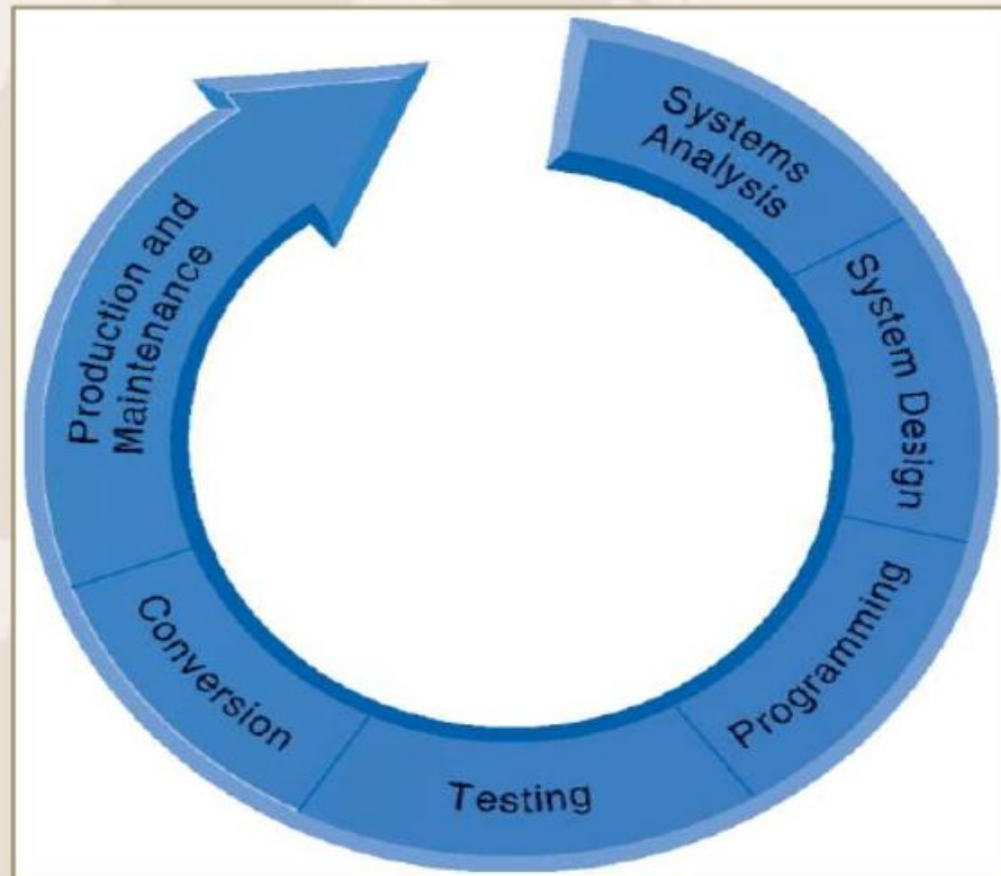
System Development

Overview of Systems Development

THE SYSTEMS DEVELOPMENT PROCESS

Building a system can be broken down into six core activities.

FIGURE 13-4



System Analysis

- **What is System Analysis?**
 - **System Analysis** is the process of examining an existing business problem or opportunity to determine the requirements for a new or improved information system.
 - It focuses on **understanding business needs before designing a technical solution.**
- **Key Activities**
 - Identify and define the business problem or opportunity.
 - Analyze the root causes of existing issues.
 - Gather and document user and business requirements.
 - Evaluate alternative solutions.
 - Prepare a **System Proposal Report** recommending the most suitable solution.
- **Output**
 - A clear and complete **System Requirements Specification (SRS)** that serves as the foundation for system design.

System Analysis

- **Establishing Information Requirements and Feasibility Analysis**
- A systems analyst determines:
 - **Who** needs the information?, **What** information is required?
 - **When** is it needed? , **Where** will it be used?
 - **How** will it be processed and delivered?
- The analyst also defines:
 - System objectives
 - Functional requirements (what the system should do)
 - Non-functional requirements (security, performance, reliability, usability)
- **Feasibility Analysis**
 - Before development begins, the proposed system is evaluated to determine whether it is practical and worthwhile.
- **Key Feasibility Dimensions**
 - **Technical:** Is the required technology and expertise available?
 - **Economic:** Do the expected benefits justify the costs?
 - **Operational:** Will users and the organization accept the new system?
 - **Schedule:** Can the project be completed within the required timeframe?

System Analysis

- **Why System Analysis Matters**

- A thorough system analysis helps organizations:
 - Clearly understand business requirements
 - Reduce project risks and development costs
 - Improve communication among stakeholders
 - Deliver systems that meet user expectations
 - Increase the likelihood of project success

- **Common Causes of Systems Failure**

- Poor or incomplete requirements gathering
- Lack of stakeholder involvement
- Inadequate feasibility analysis
- Frequent changes in requirements (scope creep)
- Weak communication between users and developers

System Design

- **What is System Design?**
 - **System Design** is the process of translating business and user requirements identified during systems analysis into a detailed blueprint for developing an information system.
 - It defines **how the system will be built** to achieve organizational objectives and satisfy user needs.
- **Major Design Activities**
 - **Database Design** – Structure and organization of data.
 - **User Interface (UI/UX) Design** – User-friendly screens and interactions.
 - **Process Design** – Business workflows and system logic.
 - **Security Design** – Access control, privacy, and cybersecurity measures.
 - **System & Network Architecture** – Hardware, software, cloud, and communication infrastructure.
- **Outcome**
 - A comprehensive **System Design Specification (SDS)** that guides system development, testing, and implementation.

System Design

- **Characteristics of Good System Design**

- A well-designed system should:
 - Meet all **functional and non-functional requirements** identified during system analysis.
 - Align **business objectives, organizational processes, and technology**.
 - Be **secure, scalable, reliable, maintainable, and user-friendly**.
 - Support future growth and changing business needs.

- **Role of End Users**

- End users play a critical role throughout the design process by:
- Validating system requirements.
- Providing feedback on prototypes and interface designs.
- Ensuring the system supports real business processes and priorities.
- Participating in user acceptance testing (UAT).

Programming

- **What is Programming?**
 - **Programming (or Coding)** is the process of translating the **System Design Specification (SDS)** into executable software using programming languages, development frameworks, and software development tools.
 - The objective is to build a system that accurately implements the functional and non-functional requirements identified during systems analysis and design.
- **Key Programming Activities**
 - Develop application modules and business logic.
 - Create and configure databases.
 - Integrate system components and external services (APIs).
 - Implement security controls and error handling.
 - Document source code for maintenance and future enhancements.
- **Output**
 - A functional software system ready for testing and validation.

System Testing

- **What is System Testing?**
 - **System testing** is the process of verifying and validating that the developed system functions correctly, meets user requirements, and is ready for deployment.
- **Levels of Testing**
 - **Unit Testing:** Tests individual modules or program components independently.
 - **Integration Testing:** Verifies that different modules work together correctly.
 - **System Testing:** Evaluates the complete integrated system against the specified requirements.
 - **User Acceptance Testing (UAT):** Conducted by end users to confirm that the system is suitable for real-world operational use.
- **Test Plan**
 - A **Test Plan** outlines:
 - Testing objectives and scope
 - Test cases and procedures
 - Required resources and responsibilities
 - Test schedule and acceptance criteria
 - Expected outcomes and defect reporting process

System Conversion

- **What is System Conversion?**
 - System Conversion is the process of replacing the existing information system with a new one and making it operational within the organization.
 - The objective is to ensure a **smooth transition** with minimal disruption to business operations.
- **Four main conversion strategies**
 - 1. Parallel strategy
 - 2. Direct cutover
 - 3. Pilot study
 - 4. Phased approach

System Conversion

- Comparison of Common Conversion Strategy

Strategy	Description	Advantages	Disadvantages
Parallel Conversion	Old and new systems operate simultaneously for a period.	Low risk; easy fallback if problems occur.	Expensive and resource-intensive.
Direct Cutover	The old system is immediately replaced by the new system.	Fast and low cost.	High risk if the new system fails.
Pilot Conversion	The new system is first implemented in one department or location before organization-wide deployment.	Problems are identified before full rollout.	Longer implementation time.
Phased Conversion	The new system is introduced gradually, module by module or department by department.	Lower risk and easier change management.	Full implementation takes longer.

System Conversion

- **Key Conversion Activities**
- **End-User Training**
 - Train users on system features, procedures, and best practices.
 - Improve user acceptance and reduce resistance to change.
- **Data Migration**
 - Transfer data from the legacy system to the new system while ensuring accuracy and integrity.
- **Documentation**
 - Prepare **Technical Documentation** for developers and IT support.
 - Prepare **User Documentation** (user manuals, online help, and training guides) for end users.
- **Post-Implementation Support**
 - Monitor system performance.
 - Resolve operational issues and user concerns.
 - Provide maintenance, updates, and continuous improvements.

Production and Maintenance

- **Production**
 - Once implemented, the information system enters the **production (operational) phase**, where it is used to support day-to-day business activities.
 - During this phase, the system is continuously monitored to ensure it:
 - Meets business and user requirements.
 - Operates reliably and securely.
 - Delivers the expected business value and performance.
- **Post-Implementation Review (PIR)**
 - A **Post-Implementation Review (PIR)** or audit is conducted to evaluate:
 - Whether project objectives have been achieved.
 - User satisfaction and system performance.
 - Costs versus expected benefits.
 - Lessons learned for future projects.
- **Outcome**
 - Recommendations for system enhancements, corrective actions, and continuous improvement.

Production and Maintenance

- **What is Maintenance?**
 - **System Maintenance** involves modifying the operational system to correct faults, adapt to changing business needs, improve performance, and enhance functionality throughout its lifecycle.
- **Major Types of Maintenance**
 - **Corrective Maintenance:** Fixes software defects and operational errors.
 - **Adaptive Maintenance:** Updates the system to accommodate changes in hardware, software, regulations, or business processes.
 - **Perfective Maintenance:** Enhances system features, usability, documentation, and processing efficiency based on user feedback.
 - **Preventive Maintenance:** Improves system reliability and security by addressing potential issues before failures occur.
- **Typical Maintenance Effort**
 - **~20%** – Bug fixes, debugging, and emergency support.
 - **~20%** – Technology upgrades (hardware, software, databases, reports, and infrastructure).
 - **~60%** – User-requested enhancements, performance optimization, security improvements, and documentation updates.

System Modeling and Design

- **System Modeling and Design**
 - System modeling provides a visual representation of an information system, helping analysts, developers, and stakeholders understand, communicate, and document system requirements and design.
- **Major Methodologies**
 - **1. Structured Methodology**
 - Follows a **step-by-step (sequential)** approach.
 - Focuses on **business processes and data flows**.
 - Treats **data** and **processes** as separate components.
 - Commonly used in traditional SDLC (Waterfall).

System Modeling and Design

– 2. Object-Oriented Development (OOD)

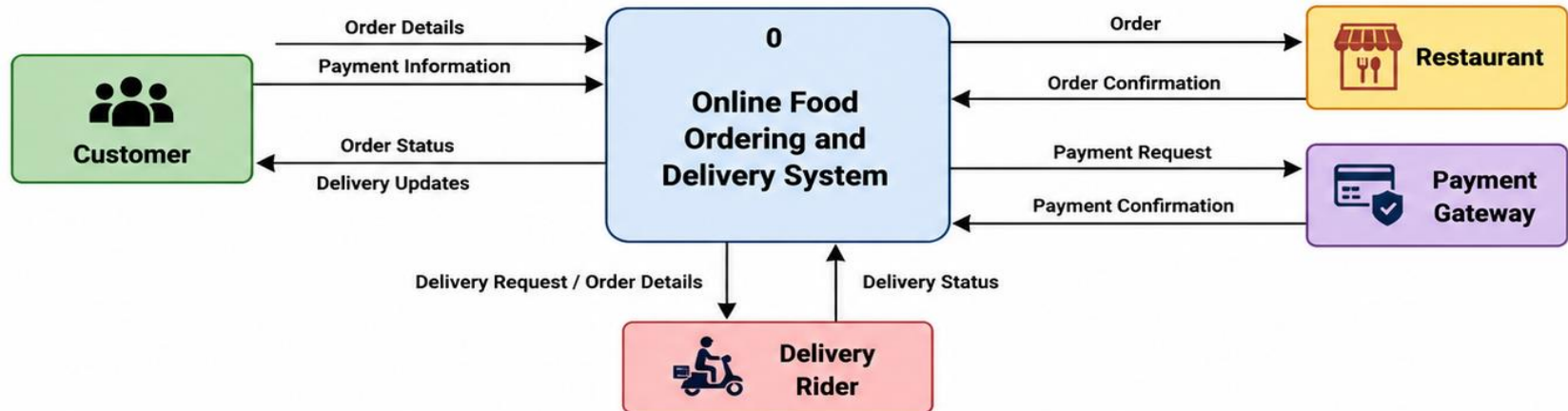
- Models the system as interacting **objects**.
 - Integrates **data and behavior** within the same object.
 - Supports **reuse, modularity, scalability, and maintainability**.
 - Widely used with **Agile development**, UML, Java, C#, and Python.
- **Key Difference:**
 - Structured methodology is **process-oriented**, while Object-Oriented Development is **object-oriented**, focusing on real-world entities.

System Modeling and Design

- **Structured Methodology and Data Flow Diagrams (DFDs)**
- **Structured Analysis**
 - Structured analysis models **how data moves through a system** and how business processes transform that data.
- **Primary Modeling Tools**
 - **Data Flow Diagram (DFD)**
 - A graphical model showing:
 - **External entities** (sources/destinations)
 - **Processes** (transform data)
 - **Data flows**
 - **Data stores**
- DFDs can be developed at different levels:
 - **Context Diagram (Level 0)** – Overall system view
 - **Level 1, Level 2, etc.** – Increasing levels of detail
- **Supporting Documentation**
 - **Data Dictionary:** Defines data elements, flows, and data stores.
 - **Process Specifications (Mini-Specs):** Describe processing logic for each process.
 - **Structure Charts:** Show the modular hierarchy and relationships among system components.

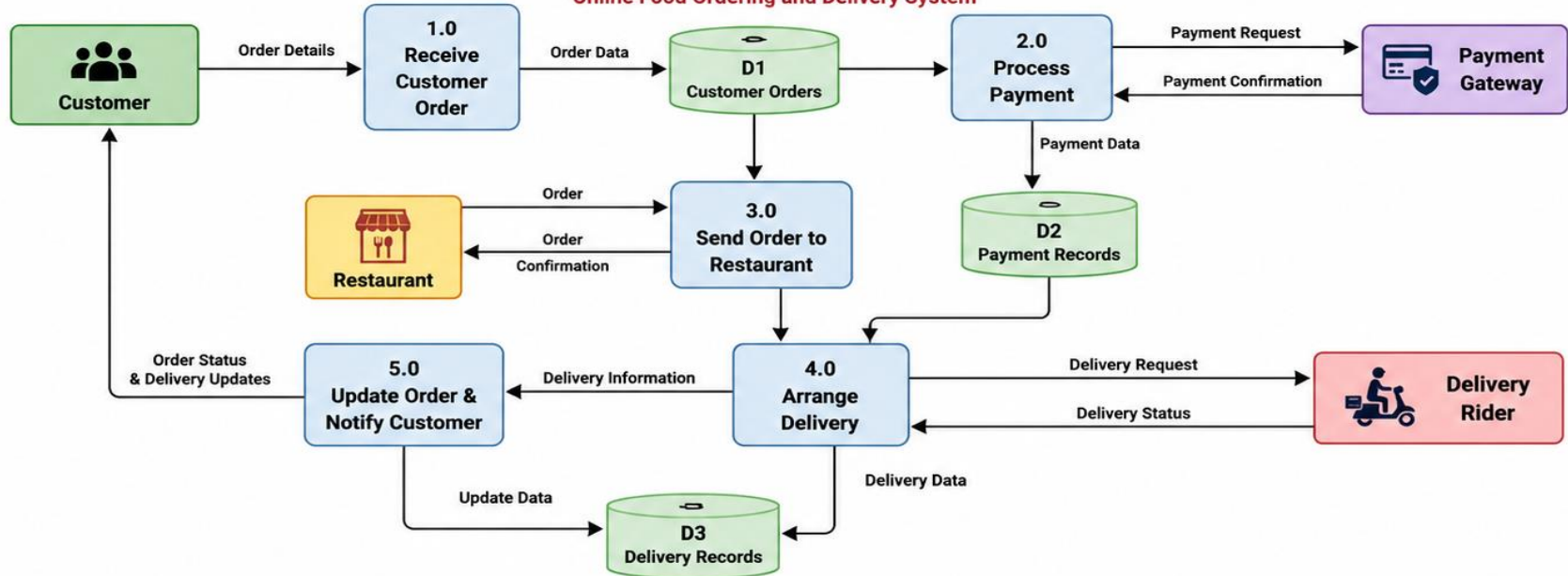
LEVEL 0 DFD (Context Diagram)

Online Food Ordering and Delivery System



LEVEL 1 DFD

Online Food Ordering and Delivery System



Explanation of Components

- **External Entities**
 - **Customer** – Places food orders and receives updates.
 - **Restaurant** – Receives and prepares orders.
 - **Payment Gateway** – Processes electronic payments.
 - **Delivery Rider** – Delivers food and updates delivery status.
- **Processes**
 - **Receive Customer Order**
 - **Process Payment**
 - **Send Order to Restaurant**
 - **Arrange Delivery**
 - **Update Order & Notify Customer**
- **Data Stores**
 - **D1 Customer Orders**
 - **D2 Payment Records**
 - **D3 Delivery Records**

System Modeling and Design

- **Object-Oriented Development (OOD)**
- **What is an Object?**
- An **object** represents a real-world entity that combines:
 - **Attributes (Data)** – Characteristics of the object.
 - **Methods (Behavior)** – Operations the object can perform.
 - Example:
 - **customer**
 - Attributes: CustomerID, Name, Delivery Address
 - Methods: PlaceOrder(), MakePayment(), TrackOrder()
- **Object-oriented modeling is based on concepts of class and inheritance**
- • Objects belong to a certain class and have features of that class
- • May inherit structures and behaviors of a more general, ancestor class
- **Modeling Language**
- The most widely used modeling language for OOD is **Unified Modeling Language (UML)**,

UML

- What is UML (Unified Modeling Language)?
 - UML is a standardized visual modeling language used in Software Engineering to analyze, design, document and communicate the structure and behavior of software systems
 - It uses diagrams and notations to represent system, rather than code.
 - It is a visual language that helps team understand and build software systems more effectively

UML

- UML applications
 - UML has several applications some of which include:
 - Planning and designing software before coding begins
 - Visualizing how a system works (structure, behavior, interaction)
 - Communicating ideas between developers, designers, and stakeholders
 - Documenting the existing system for maintenance or handover

UML Diagrams

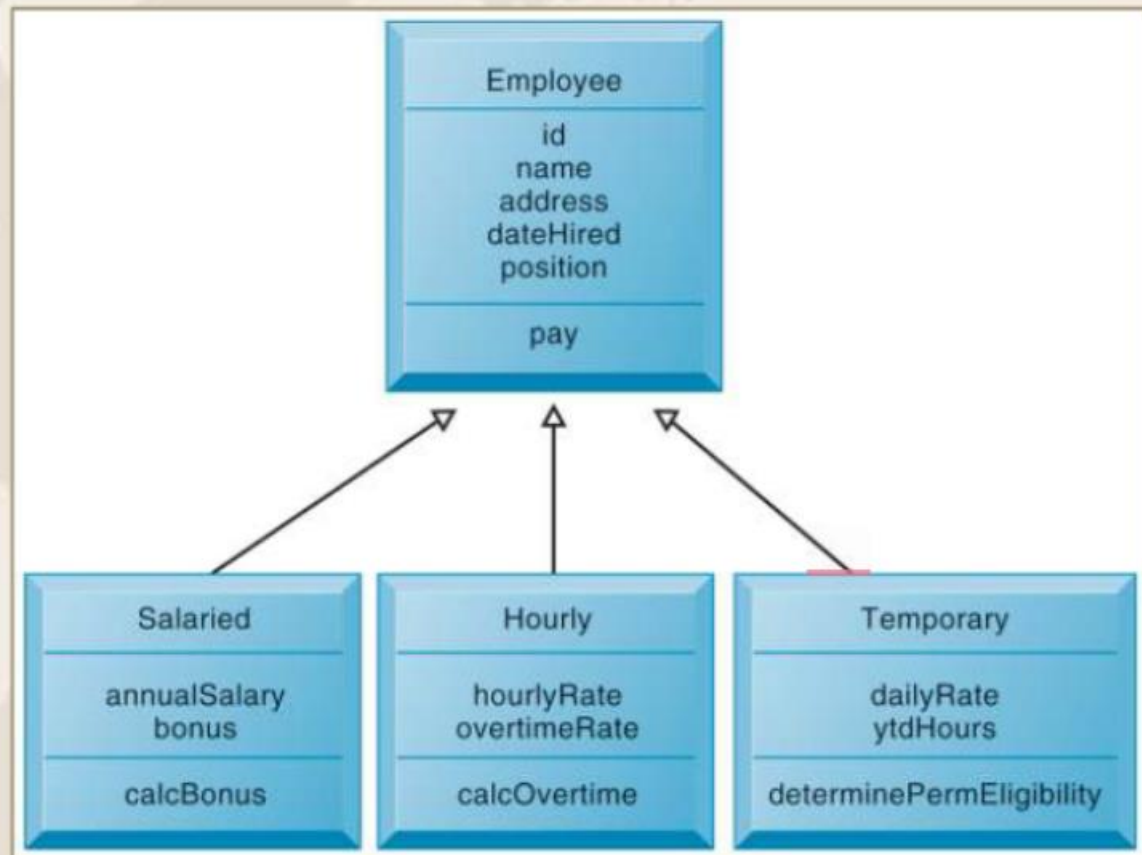
- UML has several diagrams which can be grouped into two:
 - Structural diagrams – use to represent the static structure of software system
 - E.g.
 - Class diagram
 - Object diagram
 - Component diagram
 - Etc.
- Behavioral diagram – use to show how the system behaves over time.
 - E.g.
 - use case diagram
 - Activity diagram
 - State machine
 - Etc

System Modeling and Design

CLASS AND INHERITANCE

This figure illustrates how classes inherit the common features of their superclass.

FIGURE 13-8



OBJECT-ORIENTED ANALYSIS (OOA)

Online Food Ordering and Delivery System

What is an Object?

An object represents a real-world entity that combines:

- **Attributes (Data)** – Characteristics of the object.
- **Methods (Behavior)** – Operations the object can perform.

Example Object: Customer

Attributes (Data)

- CustomerID
- FullName
- PhoneNumber
- Email
- DeliveryAddress

Methods (Behavior)

- Register()
- Login()
- BrowseMenu()
- AddToCart()
- PlaceOrder()
- MakePayment()
- TrackOrder()
- ViewOrderHistory()

Object-Oriented Modeling

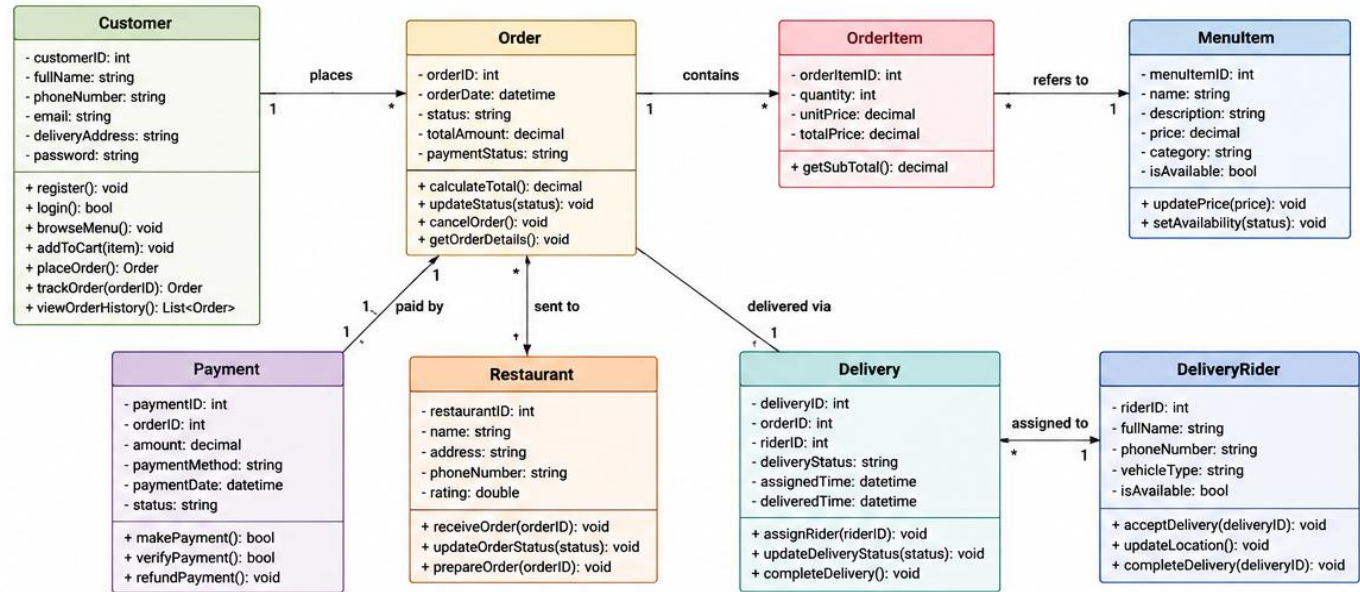
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Modeling Language

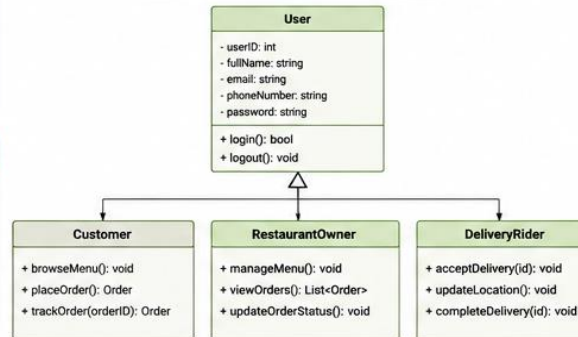
The most widely used modeling language for OOD is Unified Modeling Language (UML).



Class Diagram – Online Food Ordering and Delivery System



Inheritance (Generalization)



Key Relationships (Associations)

Relationship	Type	Description
Customer – Order	1 to Many	A customer can place many orders.
Order – OrderItem	1 to Many	An order contains one or more order items.
OrderItem – MenuItem	Many to 1	Each order item refers to one menu item.
Order – Payment	1 to 1	Each order has one payment.
Order – Restaurant	Many to 1	Many orders can go to one restaurant.
Order – Delivery	1 to 1	Each order is delivered via one delivery.
Delivery – DeliveryRider	Many to 1	Many deliveries can be assigned to one rider.

Legend

- + : Public (Method)
- : Private (Attribute)
- 1 : One
- * : Many

System Modeling and Design

- **Object-oriented development**
 - More iterative and incremental than traditional structured development
- **Systems analysis**
 - Interactions between the system and users are analyzed to identify objects
- **Design phase**
 - Describes how objects will behave and interact; grouped into classes, subclasses, and hierarchies
- **Implementation**
 - Some classes may be reused from the existing library of classes, others created or inherited
- **Because objects are reusable, object-oriented development can potentially reduce the time and cost of development**

Other methods for System Development

- **Alternative Systems-Building Methods**
 - – **Traditional systems life-cycle**
 - – **Prototyping**
 - – **End-user development**
 - – **Application software packages**
 - – **Outsourcing**

Alternative Systems-Building Methods

- **Traditional systems lifecycle:**
 - Oldest method for building information systems
 - Phased approach divides development into formal stages
 - Follows “waterfall” approach:
 - Tasks in one stage finish before another stage begins
 - **Maintains formal division of labor between end users and information systems specialists**
 - **Emphasizes formal specifications and paperwork**
 - **Still used for building large complex systems**
 - **Can be costly, time-consuming, and inflexible**

Alternative Systems-Building Methods

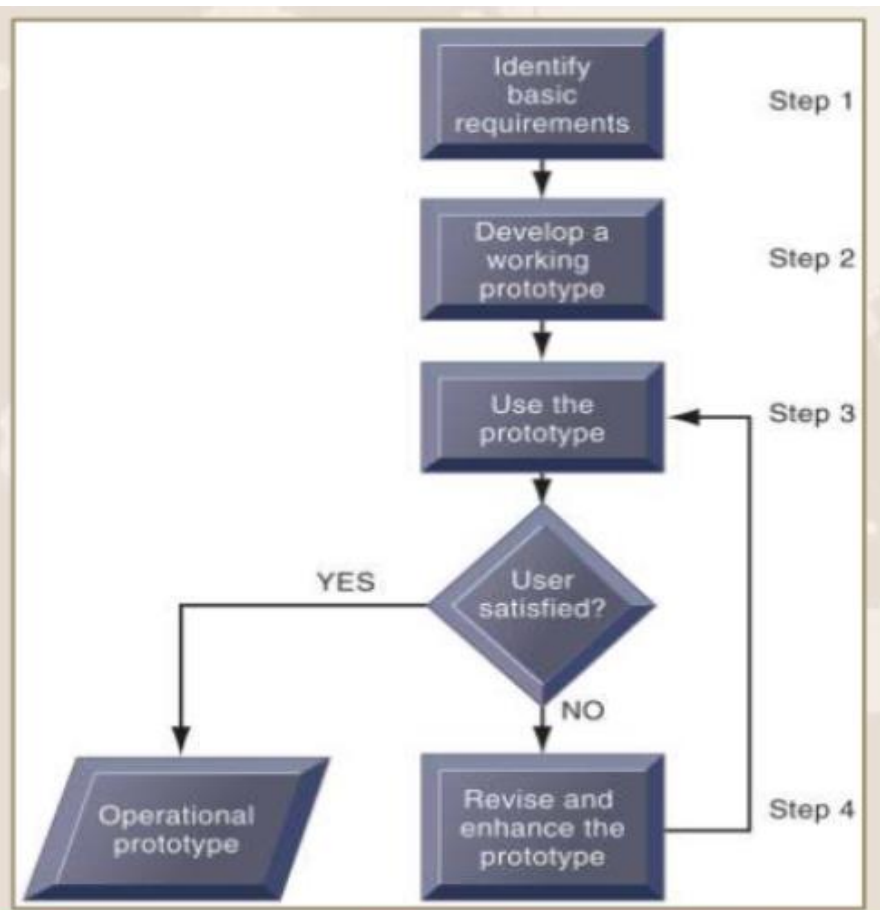
- **Prototyping**
 - Building an experimental system rapidly and Inexpensively for end users to evaluate
 - **Prototype: Working but preliminary version of Information system**
 - Approved prototype serves as template for final system
 - **Steps in prototyping**
 - 1. Identify user requirements
 - 2. Develop initial prototype
 - 3. Use prototype
 - 4. Revise and enhance the prototype

Alternative Systems-Building Methods

THE PROTOTYPING PROCESS

The process of developing a prototype can be broken down into four steps. Because a prototype can be developed quickly and inexpensively, systems builders can go through several iterations, repeating steps 3 and 4, to refine and enhance the prototype before arriving at the final operational one.

FIGURE 13-9



Alternative Systems-Building Methods

- **Advantages of prototyping**
 - Useful if some uncertainty in requirements or design solutions
 - Often used for end-user interface design
 - More likely to fulfill end-user requirements
- **Disadvantages**
 - May gloss over essential steps
 - May not accommodate large quantities of data or large number of users
 - May not undergo full testing or documentation

Alternative Systems-Building Methods

- **Application software packages**
 - **Save time and money**
 - **Many offer customization features:**
 - Software can be modified to meet unique requirements without destroying integrity of package software
 - **Evaluation criteria for systems analysis include:**
 - Functions provided by the package, flexibility, user friendliness, hardware and software resources, database requirements, installation and maintenance efforts, documentation, vendor quality, and cost
 - **Request for Proposal (RFP)**
 - Detailed list of questions submitted to packaged-software vendors
 - Used to evaluate alternative software packages